

Responsible AI Augmentation

Objective

Artificial Intelligence (AI) was incorporated throughout the project as a productivity enhancement tool to accelerate development, documentation, troubleshooting, and workflow design. AI served as a technical assistant rather than a decision making system. All analytical decisions, validation procedures, and data quality assessments remained under analyst control.

AI-Assisted Activities

Project Planning and Architecture

- Generated the overall project workplan
- Designed the Bronze, Silver, Gold data architecture
- Generated schema creation scripts for Bronze, Silver, and Gold layers

Data Ingestion

- Generated scripts for creating the raw data table
- Generated bulk insert scripts for loading source data into the Bronze layer
- Assisted in troubleshooting bulk insert issues, including:
 - Incorrect row terminators
 - Insufficient VARCHAR column lengths
 - Data loading failures

Data Profiling

- Generated a structured data profiling workflow
- Generated NULL analysis scripts using aggregate functions
- Assisted in identifying and documenting data quality dimensions

Data Standardization

- Consolidated city name variations into standardized formats
- Consolidated complaint type variations into standardized formats
- Generated mapping logic for:
 - City standardization
 - Borough standardization
 - Complaint type standardization

Data Quality Framework Development

- Generated an initial Data Quality Rulebook draft based on profiling findings

- Generated a data cleaning workflow based on the profiling results and rulebook
- Assisted in designing validation methodologies

SQL Development

- Generated CASE WHEN mapping statements for data standardization
- Generated datatype conversion scripts to transform ingestion VARCHAR columns into appropriate datatypes based on the data dictionary

Validation and Reporting

- Generated a rule validation draft using the raw data profile, cleaned data profile, and rulebook
- Generated a cleaning impact analysis draft using the raw data profile, cleaned data profile, and rulebook
- Assisted in drafting project documentation and report structure

Human Oversight

The following activities remained fully analyst owned:

- Dataset selection
- Data profiling execution
- Interpretation of profiling results
- Definition of remediation strategies
- Data cleaning decisions
- Exception handling
- Validation of cleaning outcomes
- Data quality scoring
- Final conclusions and recommendations

Data Governance Considerations

AI was used exclusively for workflow design, SQL generation, documentation assistance, troubleshooting, and productivity enhancement.

No raw operational records were shared for automated analysis. All generated outputs were reviewed, validated, and approved by the analyst prior to implementation.

This approach demonstrates how AI can be integrated into modern data workflows while maintaining accountability, validation rigor, and governance controls.